



Sri Lanka Technology Campus

CCS3440 – Artificial Intelligence

Problem Definition and Literature Review

Prediction Task Selected: Option B – Patient Readmission Prediction (30 days)

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1. Problem Definition

SmartCare Hospital maintains records of patients that are taken in and treated during normal hospital processes. As well as discharge data. An ongoing problem is that a fraction of discharged patients returns to the hospital and is admitted shortly after the first discharge. When patients can be identified early as being at increased risk of readmission, the staff can make earlier interventions (such as a follow-up call, changing the discharge plan or close monitoring of the patient), which may alleviate unhelpful readmissions.

This challenge is addressed in this coursework by formulating it as a binary classification problem. Given the demographic, clinical and admission attributes of a patient, determine whether the patient was readmitted within 30 days of discharge (`readmitted_30_days = 1`) or not (`readmitted_30_days = 0`).

2. Significance of the Problem

- Clinical significance: Early readmission is a frequent metric for the quality and comprehensiveness of care received in the acute setting in many countries.
- Financial significance: Readmissions increase the cost burden on both the hospital and the patient (additional room charges, lab charges, medicine charges as reflected in this dataset's billing fields).
- Operational significance: The prediction of risk for readmission in advance enables the hospital to consider how to manage that risk. Abilities for the management to better plan the number of beds, the number of employees and the follow up for discharge.

In the US, nearly 20% of Medicare patients are readmitted within 30 days of discharge, costing an estimated \$17 billion annually, of which about 75–76% are potentially avoidable (Jencks, Williams & Coleman, 2009; MedPAC, 2014).

3. Literature Review

In this section, we provide a summary of five research papers that were peer-reviewed and relevant to AI/ML-based hospital readmission prediction. In each paper the approach taken, important findings and limitations are identified and the approach taken in this coursework is placed in terms of existing research.

	Citation (Author, Year, Title, Venue)	AI/ML Approach Used	Key Findings	Limitation / Gap
01	Wang, H., Cui, Z., Chen, Y., Avidan, M., Abdallah, A. B., & Kronzer, A. (2018). Predicting hospital readmission via cost-sensitive deep learning. IEEE/ACM Transactions on Computational Biology and Bioinformatics (journal). https://doi.org/10.1109/TCBB.2018.2827029	Cost-sensitive deep learning for time-series vital signs: CNN + categorical feature embedding + MLP baseline, versus SVM, Logistic Regression, Decision Tree, Random Forest and ANN baselines.	The generally-ward AUC for 30-day readmission was 0.70 -- the best of all tested baselines, and is equivalent to about half the PPV of the best baseline. Implementing in an active clinical decision support system at Barnes-Jewish Hospital.	Significant class imbalance even after applying cost sensitive correction, which reduces the sensitivity of the model. Model based on vital sign monitoring data in the hospital's ICU/inpatient wards, not available in all hospital settings; model is "black-box", meaning it is not explainable.
02	Shao, D., Wang, M., Liang, S., Zhang, G., Wei, J., & Gu, H. (2024). Predicting the 30-day readmissions of congestive heart failure patients from diagnosis records. Proceedings of the 2024 3rd International Conference on Public Health and Data Science (conference). https://doi.org/10.1145/3718677.3718709	DeepCHF-RH: Deep learning model based on two decoders, one is Transformer (global) and another is CNN (local), which are fused using an MLP classifier. The above three models are compared to GSGD-ANN.	On MIMIC-IV data set (13018 CHF patients), achieved 90.27% accuracy and 97.07% AUC, which is better than the best baseline (RETAIN) by 0.84% AUC; ablation study demonstrated that both global features and local features are better than either of them alone.	The authors explicitly state that the model has no external validation on other hospitals/regions, limiting its generalizability. Is only applicable to a single disease cohort (CHF, not all-cause readmission) and is dependent on ICU derived data, which may not be representative of general ward populations.
03	da Silva, N. C., Albertini, M. K., Backes, A. R., & Pena, G. das G. (2023). Prediction of readmissions in	Classical machine learning- CART (classification and regression tree) with bagging, hyperparameters optimized with grid	30-day avoidable readmission rate 9.50%; AUC 0.79, sensitivity 76.3%, specificity 64.4% Most important predictors cancer diagnosis, ward, age,	Results may not be generalizable to other institutions, readmissions at other institutions could not be captured, EHR data may contain entry

	hospitalized children and adolescents by machine learning. Proceedings of the 38th ACM/SIGAPP Symposium on Applied Computing (conference). https://doi.org/10.1145/3555776.3577592	search with racing/ANOVA; repeated cross-validation (5-fold, 10 repetitions) with downsampling to correct for class imbalance.	admission source, surgical status, length of stay, multimorbidity, leucocytes, RBC, hemoglobin	errors.
04	Lu, Q., Nguyen, T. H., & Dou, D. (2021). Predicting patient readmission risk from medical text via knowledge graph enhanced multiview graph convolution. Proceedings of the 44th International ACM SIGIR Conference on Research and Development in Information Retrieval (conference). https://doi.org/10.1145/3404835.3463062	MedText: multiview graph convolutional network that combines intra-document, intra-UMLS and document-UMLS interactions via a knowledge-graph-enhanced document graph, decoded with attention and combined with a Bi-LSTM text encoder; compared against BioBERT, ClinicalBERT and CC-LSTM	Achieved AUROC 0.825, AUPRC 0.632, RP80 0.319 on MIMIC-III discharge summaries, outperforming all transformer-based baselines. Ablation showed all four graph views contributed to performance	Entity linking (ScispaCy) missed 15-25% of relevant clinical entities, likely limiting performance; relies only on text (single MIMIC-III dataset), no direct comparison with numerical/time-series feature models (e.g., Lin et al.'s AUROC 0.791 noted as not directly comparable)
05	Gao, W., Gao, X., & Chen, Y. (2024). Compositional and hierarchical semantic learning model for hospital readmission prediction. Proceedings of the 33rd ACM International Conference on Information and Knowledge Management (conference). https://doi.org/10.1145/3627673.3679814	CHSLM: heterogeneous medical-composition graphs (encoded via HGT) capture word-to-composition semantics. Then, a mutual BiAffine transformation connects the composition embeddings with the note-level embedding to capture composition-to-note impacts.	Outperformed all baselines on all metrics and datasets (e.g. AUROC 0.760 vs. best baseline 0.746 on discharge summaries). Ablation study showed composition-embedding module contributed most to performance (removal dropped RP80 by 20.5%).	High model complexity (multi-stage graph construction + HGT + BiAffine transformation) may hinder real-world deployability. Relies on external NLP tools (SpaCy, UMLS) fine-tuned for English clinical text, limiting portability in non-English settings. Evaluated on a single-institution dataset (MIMIC-III), thus generalizability across different hospitals is not proven.

3.1 Discussion of Existing AI Approaches

The trend observed throughout the five papers reviewed is that the machine learning models have evolved from the classical structured data models to the deep learning models and in the latest to the graph-based and hierarchical semantic models. Both Wang et al. (2018) and Shao et al. (2024) use structured/tabular EHR information (vital signs, lab results, comorbidities, etc.) with the former leveraging deep learning (CNN) and the latter deep learning (Transformer-based architectures) and da Silva et al. (2023) demonstrate that a relatively simple, interpretable ensemble method (CART with bagging) can also perform well (AUC 0.79) in the structured pediatric context. Lu et al. (2021) and Gao et al. (2024), on the other hand, abandon any structured features and rely on unstructured clinical notes instead, training the unstructured notes as graph neural networks with medical knowledge bases (UMLS), but Gao et al. model, CHSLM, is explicitly trained on and outperforms Lu et al. model, MedText. This demonstrates two literature trends (1) from tabular features to deep architectures that learn temporal and relational patterns automatically and (2) use of unstructured clinical text using transformer and graph-based NLP techniques, not just structured administrative data.

3.2 Common Algorithms Identified

- Classical ML: CART / Decision Trees with bagging (da Silva et al., 2023), best AUC 0.79.
- Convolutional Neural Networks (CNN): for time-series vital signs, local feature extraction (Wang et al., 2018; Shao et al., 2024).
- Transformer-based architectures: Transformer decoders and clinical LLMs (ClinicalBERT, BioBERT, SciBERT, CORE, BioLinkBERT, BioGPT) as standalone classifiers and embedding layers of larger models (Shao et al., 2024; Lu et al., 2021; Gao et al., 2024).
- Multilayer Perceptron (MLP): A final classification layer for multiple deep learning pipelines (Wang et al., 2018; Shao et al., 2024).
- Graph Neural Networks (GNN): heterogeneous and knowledge graph enhanced GNNs (Lu et al., 2021; Gao et al., 2024) - top-performing methods overall: for the text-based methods, with GAO et al. (2024), outperforming others with AUROC 0.760–0.826 (depending on data type).
- The top performing models were graph-based deep learning on clinical text (Gao et al., 2024), and hybrid (Transformer+CNN) models on structured data (Shao et al., 2024, AUC 0.97), indicating that the models used are mostly deep learning based and hybrid (Transformer+CNN).

The most successful methods reported in the literature (Gao et al., 2024; Shao et al., 2024) are deep learning and graph-based architectures that demand a large quantity of clinical text or time-series vital sign data. However, the data set at hand at SmartCare Hospital is structured, tabular and in a setting comparable to da Silva et al., 2023 where classical ensemble methods are still competitive (AUC 0.79). This coursework is an extension of this work and trains and compares 5 tabular algorithms which are classic in ML - Logistic Regression, Decision Tree, Random Forest, K-Nearest Neighbors and XGBoost and addresses the explainability gap identified above in the coursework through Task 07 SHAP based analysis.

3.3 Current Limitations in Existing Research

- **Limited external validation:** every reviewed paper evaluates its model on data from a single hospital or a single public database (e.g. MIMIC-III/IV, Barnes-Jewish Hospital, a single Brazilian tertiary hospital) and several authors (Shao et al., 2024; da Silva et al., 2023) explicitly flag this as a limitation to generalisability across hospitals and regions.
- **Class imbalance:** readmission datasets are consistently highly imbalanced (e.g. only 6% of GHWs cases in Wang et al., 2018; 9.5% in da Silva et al., 2023), requiring cost-sensitive loss functions or downsampling techniques, which can themselves introduce bias or reduce the effective size of the training data.
- **Lack of explainability:** most deep learning and graph-based models (Wang et al., 2018; Shao et al., 2024; Lu et al., 2021) function largely as black boxes only Gao et al. (2024) attempts to make predictions interpretable for clinical staff, through composition-level attribution of which words/phrases drove each prediction.
- **Narrow population or disease scope:** several models are restricted to a single disease cohort (e.g. congestive heart failure in Shao et al., 2024) or a single age group (pediatric patients in da Silva et al., 2023), limiting applicability to general, all-cause hospital readmission.
- **Data modality and language dependence:** NLP/graph-based approaches (Lu et al., 2021; Gao et al., 2024) depend on English-language clinical notes and external tools (UMLS, SpaCy) tuned for English text, which limits their portability to hospitals that do not produce large volumes of structured English clinical documentation.

3.4 Research Gap Analysis

The following gaps in the literature have been identified that this coursework will seek to partially fill:

1. Most reviewed studies only test one or two algorithms in great detail or compare a new deep learning algorithm to only a few baselines, rather than systematically comparison multiple algorithm families (e.g. Logistic Regression, Random Forest, XGBoost, Neural Networks) side by side under equivalent preprocessing as will be done in Task 05 as part of this coursework.
2. Explainability (e.g. SHAP/LIME) is only mentioned in a minority of the reviewed papers and rarely included in the structured-data models (Wang et al., 2018; Shao et al., 2024; da Silva et al., 2023) .However, its importance for clinical trust is highlighted in this coursework as a dedicated task (Task 07).
3. No representation of South Asian hospital contexts is found in the five papers reviewed and only the most recent two (da Silva et al., 2023; Aiello et al., 2023) are based on data from a single non-Western hospital system. It is therefore an appropriate choice of course work in the context of a Sri Lankan hospital scenario (SmartCare Hospital) which has not been sufficiently represented in the existing literature.
4. Reviewed papers were always from a single hospital and had no external validation with other hospitals which further supports the need for readmission prediction research in under-represented healthcare settings.

4. References

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- [5] W. Gao, X. Gao, and Y. Chen, "Compositional and hierarchical semantic learning model for hospital readmission prediction," in *Proc. 33rd ACM Int. Conf. Inf. Knowl. Manage.*, 2024, doi: 10.1145/3627673.3679814.